ALMA Proposal Preparation Workshop

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2.5. ALMA Mock Observations

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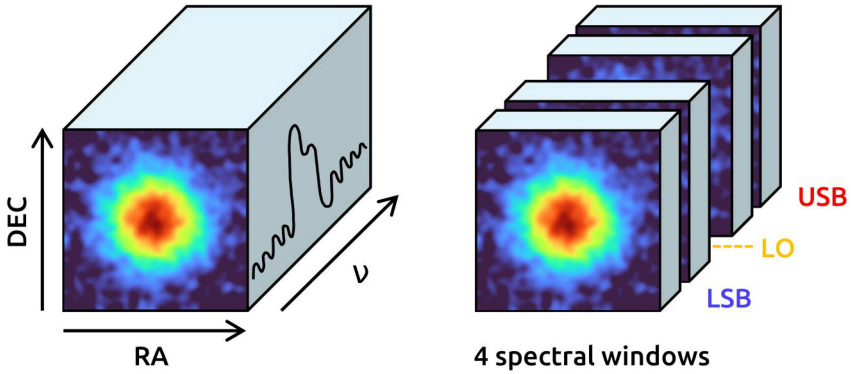
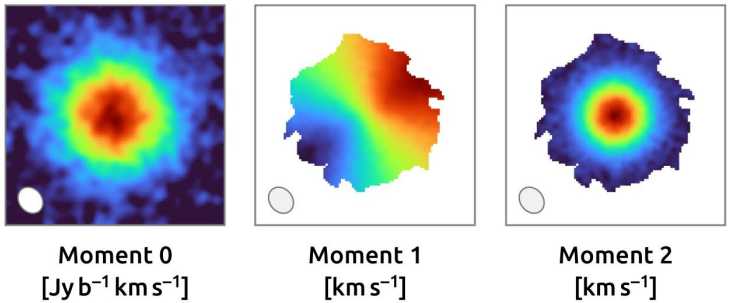
ALMA Mock Observations

- **Why simulate ALMA data?**
  - Test feasibility before submitting proposal (N_{beams}, rms sensitivity)
  - Optimize observing strategy
  - Interpret complex data (resolution effects, noise, emission filtering)
  - Compare models with observations
- **Value for proposal preparation**
  - Provide observational expectations
  - Visually justify the observing setup
  - Simulations can strengthen the proposal (often well received)





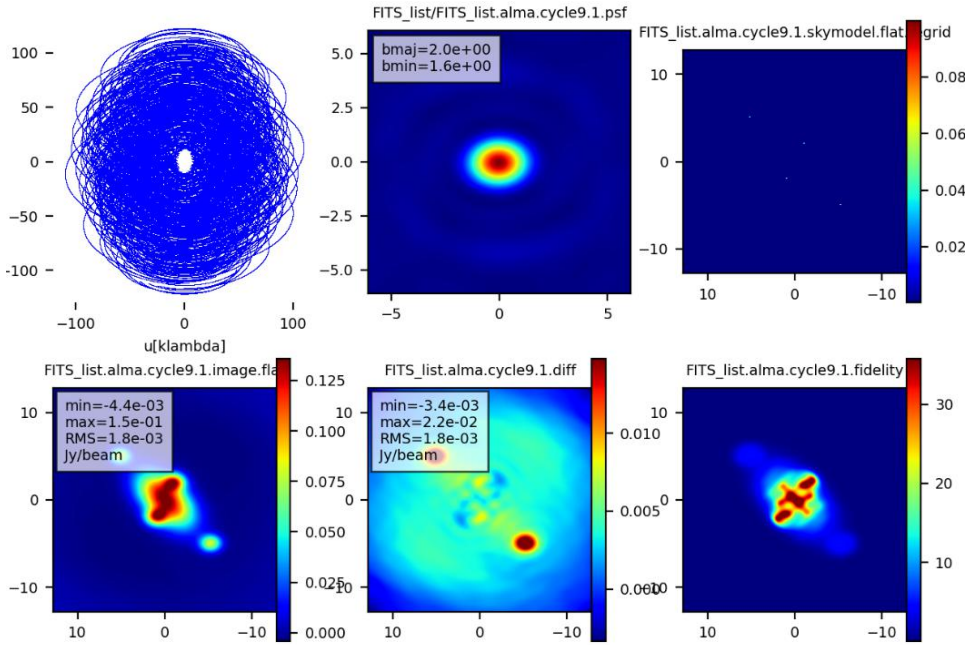
Example of mock ALMA data made for tutorials

SIMALMA: CASA Simulation Task

CASA
Common Astronomy Software Applications

CASA: calibration, imaging, simulations

Download CASA:
https://casa.nrao.edu/

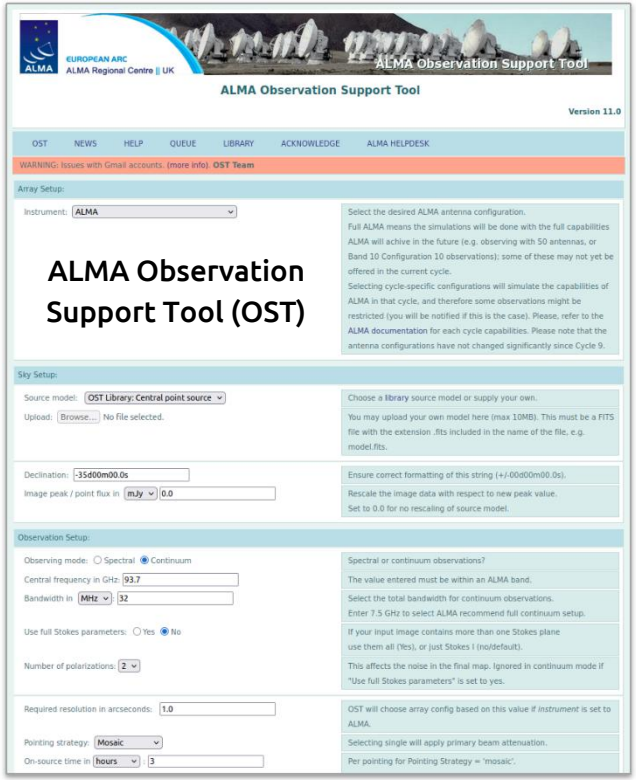


Bjorn Emonts & CASA Team

Example of SIMALMA outputs: *uv*-coverage, PSF, sky model, dirty image, residuals, fidelity

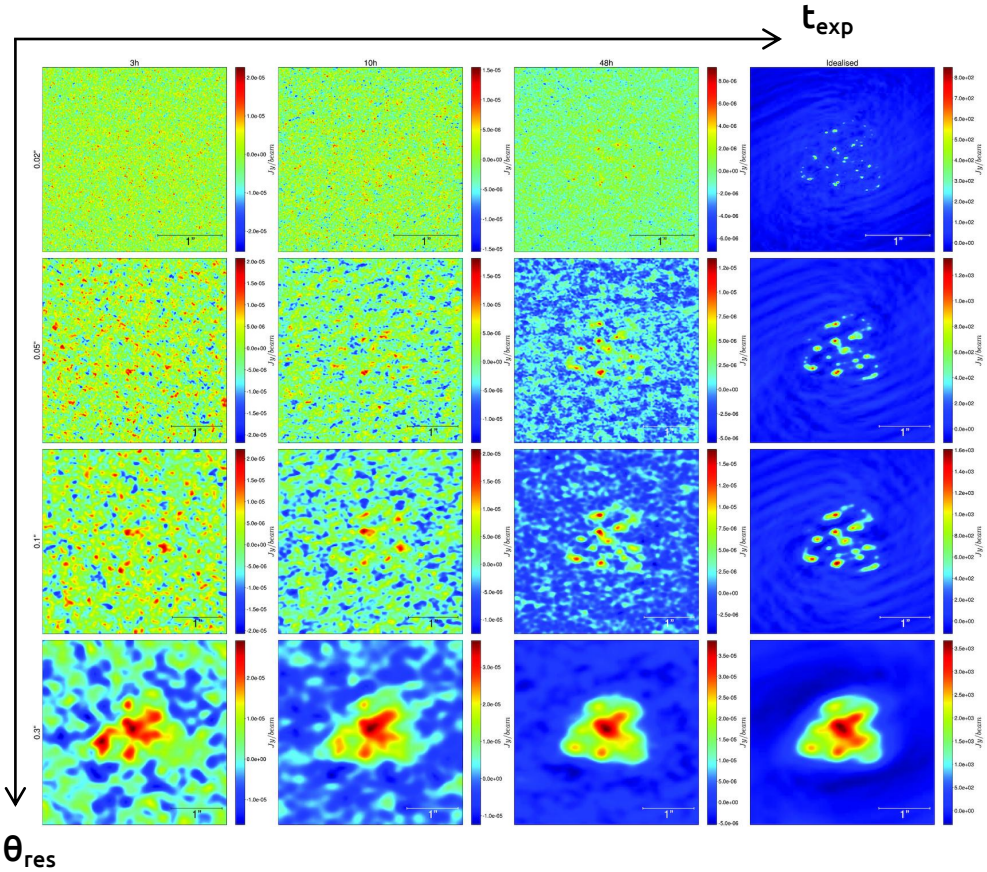
SIMALMA: https://casadocs.readthedocs.io/en/stable/api/tt/casatasks.simulation.simalma.html

ALMA Observation Support Tool (OST)



ALMA Observation Support Tool (OST)

OST: https://almaost.jb.man.ac.uk/

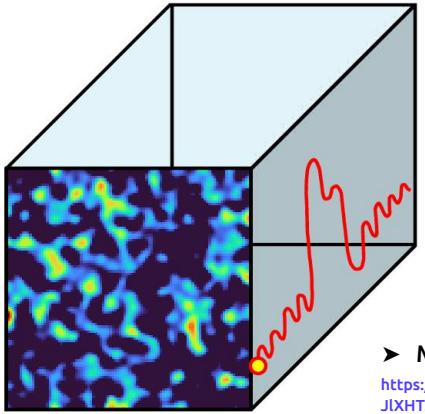


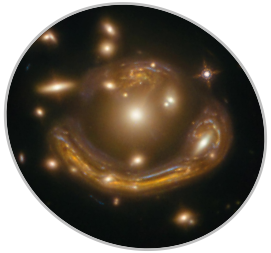
Faure et al. (2021)

Example of Mock ALMA Data using Python

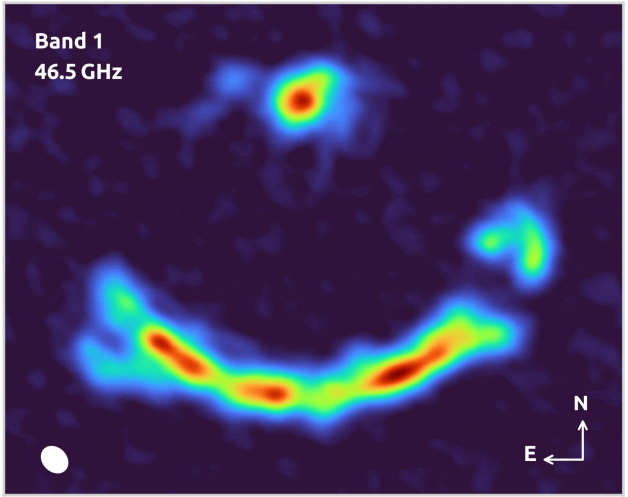
Simulation workflow (Astropy)

1) Build a sky model FITS image
2) Mimic ALMA observing parameters (beam convolution, flux normalization)
3) Generate 2D & 3D mock ALMA data





HST imaging



CO(1–0) mock imaging of the Molten Ring galaxy at z~1.5 @0.6”–1.6”

➤ Mock ALMA figures:
https://drive.google.com/drive/folders/1JlXHTZqovMDQcEMiH5vDaevE4u5Wvzz0

➤ See Jupyter notebook